Rainfall-Induced Sea Surface Emissivity Change Inferred From Satellite Passive Microwave Observations and In Situ Ship Measurements

Shaofei Wang, Zhiyong Long, Zichun Jin, Fuzhong Weng, and Yudi Liu

***Abstract*— Benefiting from the ability to penetrate clouds, microwave (MW) can theoretically achieve all-sky monitoring, thus making significant contributions to weather forecasts and climate models. Various MW sea surface emissivity (SSE) models have been applied to the products of spaceborne MW radiometers and assimilation models. However, these models do not consider the impact of rainfall on the MW SSE. In this study, rainfall-induced SSE change was inferred based on the brightness temperature (BT) observed by Advanced Microwave Scanning Radiometer 2 (AMSR2) and in situ ship measurement OceanRAIN. The neural network-based fast atmospheric parameter simulators (NN-FAPSs) were constructed for quickly calculating the atmospheric upwelling/downwelling BT and transmissivity and the errors of NN-FAPSs fitting on the SSE were below 0.001 ± 0.001 and 0.01 ± 0.03 at 6.925 and 36.5 GHz, respectively. The SSE estimates show that rainfall causes an increase of SSE (e.g., up to 0.2 at 10.65 GHz), and this phenomenon is more pronounced at low wind speeds (WSs) and high sea surface temperature (SST), corresponding to the effects of rainfall-induced local wind and SST on the dielectric constant. The error analysis shows that at 6.925, 7.3, and 10.65 GHz, the accuracy of the SSE is less affected by the accuracy of the input parameters as well as the fitting errors of the NN-FAPSs ($\Delta\varepsilon$ was below 0.01 ± 0.01). Additionally, polynomial models for 6.925, 7.3, and 10.65 GHz are proposed to provide the first guess of SSE under rainfall conditions for improving the accuracy of cloud and rainfall products. It is believed that this work helps improve the understanding of rainfall-induced MW SSE changes.**

***Index Terms*— All-sky, atmospheric correction, passive microwave (MW), rainfall, sea surface emissivity (SSE).**

I. Introduction

SEA surface emissivity (SSE) is an important parameter of the ocean–atmosphere interface and plays a fundamental role in data assimilation, numerical weather prediction, and various climate models [1], [2], [3]. Accurate SSE is crucial for interpreting satellite-based passive microwave (MW) observations and is important prior knowledge for accurate measurement of ocean environment parameters, such as sea surface temperature (SST), sea surface wind speed (WS), and precipitation [4], [5], [6], [7], [8], [9], [10]. Therefore, the community has long been committed to developing accurate and fast MW SSE models.

Currently, the methods for calculating MW SSE can be divided into two groups: radiative transfer equation (RTE)-based inversion model and physical model. The RTE-based inversion model calculates emissivity by solving various parameters in the RTE, and this method is widely used in estimating land-surface emissivity. Meissner and Wentz [11] modeled SSE between 6 and 90 GHz based on the RTE-based inversion model for a wide range of WSs and Earth incidence angles. To assimilate the radiance of the Advanced Microwave Sounding Unit [12] and the Special Sensor Microwave Imager/Sounder humidity-sounding channels [13] over land, the surface emissivities retrieved from satellite observations for the surface-sensitive channels were applied to the nearest sounding channels. Here, the physical model includes the geometric optics (GOs) model, the two-scale model (TSM), and the fast SSE model established from the result of GO model and TSM. In TSM, the sea surface roughness is described by combining the effects of small-scale capillary waves (Bragg scattering) and large-scale gravity waves [14]. For the GO model, only large-scale gravity waves are accounted and small-scale roughness is ignored [15]. Liu et al. [16] proposed an improved FAST MW Emissivity Model (FASTEM-4) based on TSM, which extended the frequency range of FASTEM from 20–60 to 1.4–410 GHz. Prigent et al. [15] constructed a fast SSE model (i.e., the tool to estimate sea surface emissivity from MWs to submillimeter waves, TESSEM^2) for the frequency range of 10–700 GHz, extending the frequency from MW to submillimeter waves. The SSE of the TESSEM^2 below 200 GHz was trained based on FASTEM, and above 200 GHz was trained by a GO model. Dinnat et al. [17] developed the passive and active reference MW to infrared ocean (PARMIO) model that computes the SSE from MW (∼0.5 GHz) to the infrared (∼400 nm). Based on this model, Kilic et al. [18] used a neural network (NN) to construct a fast emissivity model (surface fast emissivity model for ocean, SURFEM-Ocean), which has been

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implemented in the radiative transfer for TOVS (RTTOV) [19] and is considered to replace the existing FASTEM model as the default SSE model of RTTOV in the future.

Although the TSM has achieved great success in calculating MW SSE, it should be noted that TSM mainly considers the sea surface roughness caused by the wind, resulting in a nonnegligible error when used under rainfall conditions [20]. On the one hand, raindrops hitting the sea surface cause changes in the sea surface roughness spectrum induced by sea surface wind [21], [22], [23]. Ring waves generated during heavy rain can increase sea surface roughness and suppress longer gravity waves [24]. On the other hand, the downdrafts caused by falling raindrops are horizontally deflected at the sea surface resulting in local winds (WS_{local}) [25], [26]. To address this problem, the terminal velocity of the falling raindrops inferred from the raindrop diameter can be used to calculate WS_{local} based on the momentum conservation equations [27]. However, a comparison result based on measurements showed that the modeled WS_{local} still had nonnegligible errors [25]. Jin et al. [26] analyzed the impact of rainfall on L-band satellite observations by considering rainfall-induced water refreshing, ring wave, and WS_{local}, and the results showed that the rain effects on sea surface emissions were higher than 3. Ye and Guo [28] proposed a new sea spectrum by combining wind-driven spectrum and some correction factors caused by rain. With the new spectrum, the simulation error for vertical (V) polarization and horizontal (H) polarization decreased from 2.574 to 1.920 and from 3.2965 to 2.8153 dB, respectively, compared with the observational data of HY-2B scatterometer.

As far as we know, there is no all-sky SSE model covering a wide frequency range so far. The reason is that the current understanding of the sea surface microphysical parameters changes caused by rainfall is not mature enough, e.g., how to model WS_{local}, which greatly limits the construction of a physical-based all-sky SSE model. Therefore, in this study, the RTE-based inversion method was used to calculate the MW SSE under rainfall conditions. At the same time, the trend of MW SSE changes caused by rainfall with rain rate (RR), WS, and SST were analyzed. We believe that this study helps improve the understanding of rainfall-induced MW SSE changes.

II. Datasets

A. OceanRAIN

The ocean rainfall and ice-phase precipitation measurement network (OceanRAIN Version 2.0) provides in situ along-track ship-based data of 1-min resolution precipitation over the global ocean from June 2010 to December 2018 [29], [30]. The optical disdrometer ODM470 mounted on the ships is specifically designed for all-weather shipboard operations and includes a near-infrared light-emitting diode and a photodiode receiver [31]. The OceanRAIN dataset provides more than 11.6 million measurements from eight ships, and each measurement contains rain probability, intensity, and microphysical parameters. Here, the ODM470 precipitation rate was used. Additionally, sea surface and near-surface atmospheric parameters are also provided. More details about the OceanRAIN data postprocessing and quality checking can be found in [29]. The OceanRAIN dataset can be downloaded from https://opendata.dwd.de/climate_environment/CDC/.

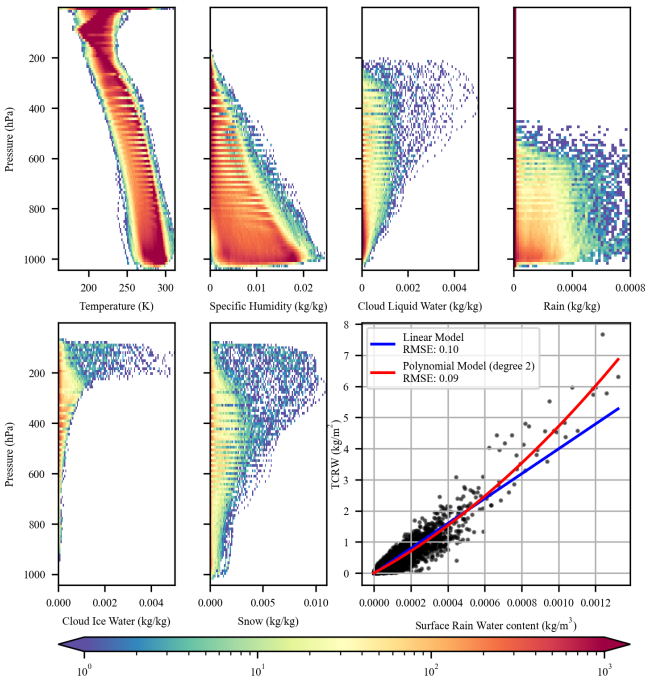

Fig. 1. Distributions of atmospheric temperature, specific humidity, cloud liquid water, rain, cloud ice water, and snow with pressure. The bottom-right subplot shows the scatterplot and regression relationship between rainwater content at the surface and TCRW.

B. Diverse Profile Dataset and Reanalysis

In this study, the diverse profile dataset [32], including a representative collection of 25 000 atmospheric profiles from the European Centre for Medium-Range Weather Forecasts (ECMWF) Integrated Forecasting System short-range forecasts, was used as the inputs of RTTOV to generate training data for the fast atmospheric parameter simulators (Section III-B). Each profile contains all the parameters used for the all-sky radiative transfer simulation and has 137 hybrid pressure levels in the vertical from the surface to ~80 km. In this study, the profiles over the land and profiles with SSTs less than 273.15 K were removed. The pressure level variables used include temperature, humidity, cloud condensate (cloud liquid water and cloud ice water), and precipitation (rain and snow). The water contents (kg/m^3) on different pressure levels were integrated over height to obtain the corresponding total column water content (TCWC, kg/m^2). The unit conversion details can be found in [33]. Fig. 1 shows the distributions of atmospheric variables used with pressure. In addition, a second-order polynomial regression relationship was established between rainwater content at the surface and total column rainwater (TCRW, kg/m^2). The single-level variables, including SST and WS, were used to estimate the SSE. The ECMWF diverse profile dataset was downloaded from https://nwp-saf.eumetsat.int/site/software/atmospheric-profile-data.

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ERA5 is the fifth generation of global atmospheric reanalysis data developed by the ECMWF [34]. The data assimilation system used to produce ERA5 is the Integrated Forecasting System Cycle 41r2 4D-Var. It provides detailed estimates of atmospheric, ocean-wave, and land-surface variables on a global scale. The hourly TCWCs except TCRW, i.e., total column water vapor (TCWV), total column cloud liquid water (TCLW), total column cloud ice water (TCIW), and total column snow water (TCSW), were extracted from the ERA5 single-level products with a grid size of 0.25°×0.25°. The ERA5 single-level products were downloaded from https://cds.climate.copernicus.eu/.

C. AMSR2 Data

The Advanced Microwave Scanning Radiometer 2 (AMSR2) onboard the Global Change Observation Mission 1st-Water satellite is a dual-polarization passive MW radiometer for measuring the MW emission from the Earth's surface and atmosphere at 6.925, 7.3, 10.65, 18.7, 23.8, 36.5, and 89.0 GHz. AMSR2 operates at an orbital altitude of about 700 km, and the antenna beam has an off-nadir angle of 47.5° (incident angle of 55° at the Earth's surface). The native spatial resolutions of AMSR2 are 35 × 62 and 3 × 5 km at 6.925 and 89.0 GHz, respectively. In this study, the 0.1° AMSR2 L3 brightness temperatures (BTs) produced by the Japan Aerospace Exploration Agency were used. The AMSR2 L3 products were downloaded at https://gportal.jaxa.jp/gpr/.

III. Theory and Method

A. Radiative Transfer Equation

Assuming a flat and specular surface, the BT observed by a passive MW sensor with the Rayleigh–Jeans approximation for an atmosphere with scattering can be expressed as [35]

$$\mathrm{BT} = \varepsilon T_s \Gamma + (1-\varepsilon) T^{\downarrow} \Gamma + T^{\uparrow} \tag{1}$$

with

$$\begin{aligned} T^{\uparrow} &= \int_0^{\tau} \left\{ \left[1-\omega(\tau')\right] T(\tau') + \omega(\tau') T_{\mathrm{sca}}(\tau') \right\} \\ &\quad \times \exp\left[-(\tau-\tau')/\mu\right] d\tau'/\mu \\ T^{\downarrow} &= \int_0^{\tau} \left\{ \left[1-\omega(\tau')\right] T(\tau') + \omega(\tau') T_{\mathrm{sca}}(\tau') \right\} \\ &\quad \times \exp\left(-\tau'/\mu\right) d\tau'/\mu + T_{\mathrm{cosm}} e^{-\tau/\mu} \\ \Gamma &= \exp\left[-\int_0^{z} \kappa_{\mathrm{ext}}(z') dz' \right] \\ T_{\mathrm{sca}}(\tau') &= B^{-1}\left[\frac{1}{4\pi} \int_0^{2\pi} \int_{-1}^{1} L(\tau';\mu';\varphi') P(\mu;\varphi;\mu';\varphi') d\mu' d\varphi' \right] \end{aligned} \tag{2}$$

where μ is the cosine of the Earth incident angle θ; ε is the surface emissivity; T_s, $T^{\uparrow}$, $T^{\downarrow}$, and T_{cosm} are the surface temperature, atmospheric upwelling BT, atmospheric downwelling BT, and cosmic background BT in K, respectively; Γ is the atmospheric transmissivity; τ is the optical thickness of the atmosphere, including absorption and scattering; ω is single scattering albedo; $T(\tau')$ and $T_{\mathrm{sca}}(\tau')$ are the atmosphere temperature and the equivalent BT of the scattering source term at τ', respectively; κ_{ext} is the atmospheric extinction at altitude z'; B^{-1} is the inverse Planck function; φ is the azimuth angle; L is the radiance at the incident position (τ', μ', and φ'); and P is the scattering phase function. Compared with clear-sky RTE, the scattering source term is added to $T^{\uparrow}$ and $T^{\downarrow}$, and the absorption coefficient is replaced by the extinction coefficient. In addition, it should be noted that as the frequency increases, the Rayleigh-Jeans approximation can introduce nonnegligible errors. Therefore, an additional Planck correction term was introduced here

$$\mathrm{BT} = \varepsilon T_s \Gamma + (1-\varepsilon) T^{\downarrow} \Gamma + T^{\uparrow} + T_{\mathrm{correct}} \tag{3}$$

where T_{correct} is the Planck correction term in K. Thus, the surface emissivity can be estimated with

$$\varepsilon = \frac{\mathrm{BT} - (T^{\uparrow} + T^{\downarrow}\Gamma + T_{\mathrm{correct}})}{(T_s - T^{\downarrow})\Gamma}. \tag{4}$$

B. Fast Atmospheric Parameter Simulators

According to (4), the key to retrieving emissivity is the calculation of three atmospheric parameters, i.e., $T^{\uparrow}$, $T^{\downarrow}$, and Γ. The radiative transfer model, e.g., RTTOV [19], Community Radiative Transfer Model [36], and Advanced Radiative Transfer Modeling System [37], relies on vertically layered atmospheric profiles that are difficult to obtain. An alternative approach is to develop atmospheric parameter calculation models with the total column content as input. In this study, the NN-based fast atmospheric parameter simulators (NN-FAPSs) were constructed for the three atmospheric parameters. Considering that the influences of the atmosphere on MW radiation mainly include the absorption and emission of water vapor and the absorption, emission, and scattering of solid and liquid water, the input parameters of the NN model included TCWV, TCLW, TCRW, TCIW, and TCSW. In addition, near-surface air temperature (NSAT) was also used as an input because it has good consistency with atmospheric parameters under high temperature and low humidity conditions [38]. The output of the NN model can be expressed as

$$\begin{aligned} y &= \mathbf{W_3} \cdot \mathrm{Relu}(\mathbf{W_2} \cdot \mathrm{Relu}(\mathbf{W_1 x} + \mathbf{b_1}) + \mathbf{b_2}) + b_3 \\ \mathrm{Relu}(x) &= \max(0, x) \end{aligned} \tag{5}$$

with

$$\begin{cases} \mathbf{x} \in \mathbb{R}^6 \\ \mathbf{W_1} \in \mathbb{R}^{128\times 6}, \mathbf{b_1} \in \mathbb{R}^{128} \\ \mathbf{W_2} \in \mathbb{R}^{256\times 128}, \mathbf{b_2} \in \mathbb{R}^{256} \\ \mathbf{W_3} \in \mathbb{R}^{1\times 256}, b_3 \in \mathbb{R} \end{cases} \tag{6}$$

where y is the estimated atmospheric parameter, $\mathbf{x}$ is the input vector, $\mathbf{W_1}$–$\mathbf{W_3}$ are the weight matrices for each layer, and $\mathbf{b_1}$–b_3 are the bias terms. The emissivity error due only to the fitting errors of the NN-FAPSs and the Planck correction term can be expressed as

$$\Delta\varepsilon_{\mathrm{fitting}} = \sqrt{\begin{array}{c} \left(\frac{\partial\varepsilon}{\partial T^{\uparrow}} \cdot \Delta T^{\uparrow}\right)^2 + \left(\frac{\partial\varepsilon}{\partial T^{\downarrow}} \cdot \Delta T^{\downarrow}\right)^2 \\ + \left(\frac{\partial\varepsilon}{\partial\Gamma} \cdot \Delta\Gamma\right)^2 + \left(\frac{\partial\varepsilon}{\partial T_{\mathrm{correct}}} \cdot \Delta T_{\mathrm{correct}}\right)^2 \end{array}} \tag{7}$$

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with

$$\frac{\partial \varepsilon}{\partial T^{\uparrow}} = -\frac{1}{(T_s - T^{\downarrow})\Gamma} \qquad \frac{\partial \varepsilon}{\partial T^{\downarrow}} = \frac{\varepsilon - 1}{T_s - T^{\downarrow}}$$
$$\frac{\partial \varepsilon}{\partial \Gamma} = -\frac{\mathrm{BT} - T^{\uparrow} - T_{\mathrm{correct}}}{(T_s - T^{\downarrow})\Gamma^2} \qquad \frac{\partial \varepsilon}{\partial T_{\mathrm{correct}}} = -\frac{1}{(T_s - T^{\downarrow})\Gamma} \tag{8}$$

where $\Delta T^{\uparrow}$, $\Delta T^{\downarrow}$, $\Delta\Gamma$, and $\Delta T_{\mathrm{correct}}$ are the fitting errors of $T^{\uparrow}$, $T^{\downarrow}$, Γ, and Planck correction term, respectively. Then, considering the errors of the NN-FAPS input parameters, BT of MW sensor, and SST, the SSE error on the whole samples can be written as

$$\overline{\Delta\varepsilon} = \overline{\Delta\varepsilon_{\mathrm{fitting}}} + \overline{\Delta\varepsilon(\Delta T^{\uparrow}_{\Delta\mathbf{x}})} + \overline{\Delta\varepsilon(\Delta T^{\downarrow}_{\Delta\mathbf{x}})} + \overline{\Delta\varepsilon(\Delta\Gamma_{\Delta\mathbf{x}})} + \overline{\Delta\varepsilon(\Delta\mathrm{BT})} + \overline{\Delta\varepsilon(\Delta T_s)}$$
$$\Delta\varepsilon_{\mathrm{std}} = \begin{pmatrix} \Delta\varepsilon^2_{\mathrm{fitting,std}} + \Delta\varepsilon_{\mathrm{std}}(\Delta T^{\uparrow}_{\Delta\mathbf{x}})^2 + \Delta\varepsilon_{\mathrm{std}}(\Delta T^{\downarrow}_{\Delta\mathbf{x}})^2 \\ +\Delta\varepsilon_{\mathrm{std}}(\Delta\Gamma_{\Delta\mathbf{x}})^2 + \Delta\varepsilon_{\mathrm{std}}(\Delta\mathrm{BT})^2 + \Delta\varepsilon_{\mathrm{std}}(\Delta T_s)^2 \end{pmatrix}^{1/2} \tag{9}$$

where $\Delta\varepsilon(\Delta T^{\uparrow}_{\Delta\mathbf{x}})$, $\Delta\varepsilon(\Delta T^{\downarrow}_{\Delta\mathbf{x}})$, and $\Delta\varepsilon(\Delta\Gamma_{\Delta\mathbf{x}})$ are the SSE errors induced by the errors of $T^{\uparrow}$, $T^{\downarrow}$, and Γ caused by the input vector $\mathbf{x}$, respectively; $\Delta\varepsilon(\Delta\mathrm{BT})$ and $\Delta\varepsilon(\Delta T_s)$ are SSE errors induced by the errors of BT of MW sensor and SST, respectively; and subscript std is the standard deviation. The SSE error analyses are shown in Section VI.

C. OceanRAIN Track-to-Area Adjustments

There is a spatial representation challenge when using the in situ ship-based OceanRAIN dataset for spatial–temporal matching with MW pixels [30]. Therefore, the precipitation parameters along-track averaged need adjustments to obtain the corresponding parameters at the MW pixel scale. Burdanowitz et al. [39], [40] proposed two statistical adjustments to address this problem derived from island-based radar data, and the root-mean-square error (RMSE) between the adjusted along-track precipitation rate and the MW pixel scale precipitation rate is reduced to 0.19 mm/h. In this study, these two adjustments were used to obtain the precipitation rate on the MW pixel scale

$$R^{*}_{\mathrm{O}} = R_{\mathrm{O}} \cdot [9.32 \cdot (T_E)^{-2.14} + 0.48]$$
$$R^{**}_{\mathrm{O}} = R^{*}_{\mathrm{O}} \cdot [0.731 \cdot (R^{*}_{\mathrm{O}}/0.18)^{-0.789} + 0.306] \tag{10}$$

where R_{O} is the along-track averaged RR (mm/h) measured by the ODM470; R^{*}_{O} and R^{**}_{O} are the RRs after the first and the second statistical adjustment, respectively; and T_{E} is the average duration of precipitation events in minutes. Hereafter, the pixel-effective RR (PERR) was used to denote R^{**}_{O}.

D. Implementation

In this study, the six frequencies (i.e., 6.925, 7.3, 10.65, 18.7, 23.8, and 36.5 GHz) from AMSR2 were selected to test the SSE changes induced by the rainfall. The reason for not including 89.0 GHz is that the effect of the spherical ice particle assumption on the simulation results can no longer be completely ignored at 89.0 GHz [41]. The implementation process of SSE inversion is shown in Fig. 2, and three steps were included.

Step I: Preprocessing of ECMWF diverse profile dataset and simulation of atmospheric radiative transfer with scattering, including separation of the profiles over the ocean from raw data, unit conversion, and calculation of TCWC, and simulation of AMSR2 BT, $T^{\uparrow}$, $T^{\downarrow}$, and Γ of the selected six frequencies with RTTOV-SCATT. In this step, the SSE was calculated by the SURFEM-Ocean. Note that the cloud fraction was set to 1 to ignore the heterogeneity of the clouds within the subgrid [33]. In addition, $T^{\uparrow}$, $T^{\downarrow}$, and Γ were directly derived from the simulation results of RTTOV-SCATT, rather than calculated by us with (2).

Step II: Construction of NN-FAPSs, including partitioning of training and validation sets, determination of Planck correction terms, model training and tuning, and calculation of Shapley additive explanations (SHAP) values [42]; 80% of BT simulation results from Step I were used for training ($N = 12\,535$) and the other 20% were used for validation ($N = 3134$). The constructed NN contains two hidden layers and one output layer. The number of neurons in the two hidden layers is 128 and 256 with the rectified linear unit (Relu) activation function. The number of neurons in the output layer is 1 with the linear activation function. SHAP is based on the Shapley value (from cooperative game theory), which treats model predictions as the result of cooperation among multiple features. The predicted value of the model is decomposed into the sum of the baseline prediction (e.g., the full sample mean) and the SHAP value of each feature. Therefore, SHAP values were calculated to explain how NN-FAPS predicts the corresponding atmospheric parameters.

Step III (SSE Retrieval Under Rainfall Conditions): In this step, the SST, WS, wind direction, salinity, PERR, and NSAT were from OceanRAIN. The other TCWCs except TCRW were from ERA5. During the spatial–temporal matching process, the time window threshold between OceanRAIN and AMSR2 was set to 20 min. The following AMSR2 pixels were not considered: 1) the number of OceanRAIN samples is less than 10; 2) all OceanRAIN samples within the AMSR2 pixel are nonprecipitation; and 3) there is a nonpure rain event within the AMSR2 pixel. This third criterion is to ensure that there is no heavy snowfall event within the selected AMSR2 pixels. Then, the ODM470 precipitation rates of all OceanRAIN samples within the same AMSR2 pixel were adjusted based on (10) after along-track averaging to obtain the corresponding PERR. The hourly TCWCs except TCRW from ERA5 were interpolated to the overpass time of AMSR2. Finally, the SSE under rainfall conditions can be calculated based on (4).

IV. NN-Based Fast Atmospheric Parameter Simulator

In this study, 36 NN-FAPS models (6 frequencies × 2 polarizations × 3 atmospheric parameters) were constructed in total, and the performances of the models are shown in Table I. It can be found that the biases and RMSEs of NN-FAPSs on the validation dataset and training dataset constructed based on RTTOV simulation results are very close, which indicates that

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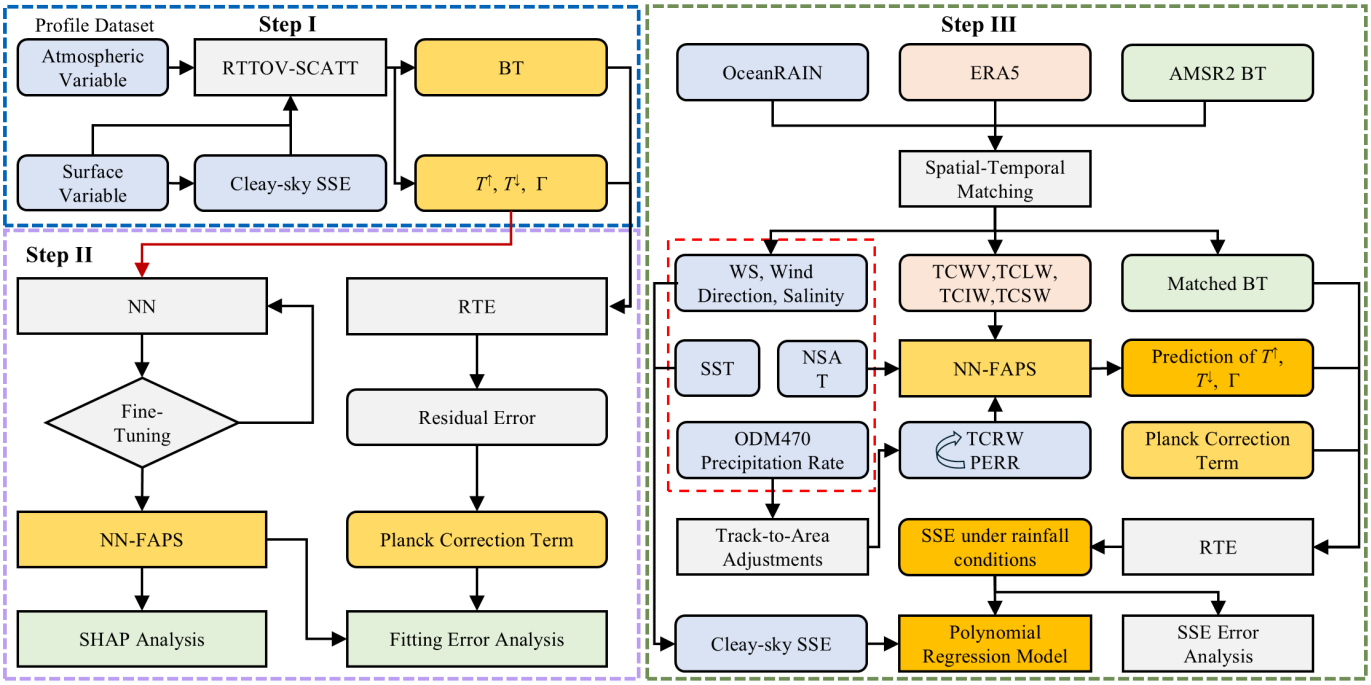

Fig. 2. Flowchart of the three steps in this study.

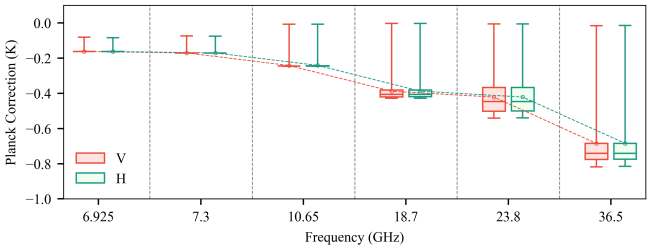

Fig. 3. Planck correction term for the selected frequencies. The box represents the data from the first quartile to the third quartile. The line in the box is the median. The circle represents the mean. The whiskers denote the 5% and 95% values.

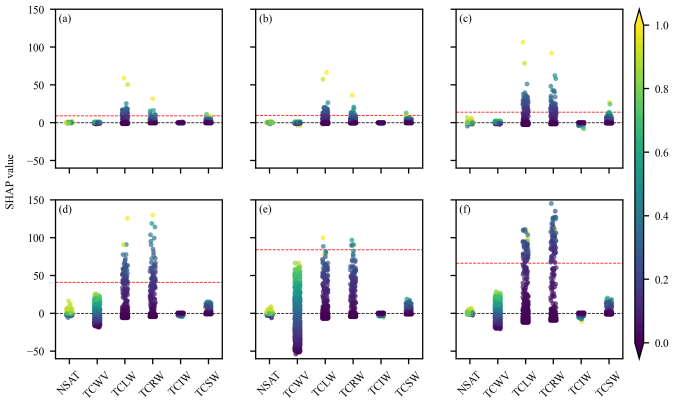

Fig. 4. SHAP values of downwelling BT at V polarization for different frequencies. The color bar represents the normalized feature value. The red dashed line represents the baseline value of NN-FAPS (i.e., model outputs when all input parameters are 0). (a) 6.925, (b) 7.3, (c) 10.65, (d) 18.7, (e) 23.8, and (f) 36.5 GHz.

no overfitting occurs. For $T^{\uparrow}/T^{\downarrow}$ and Γ, the maximum values of the bias do not exceed 0.3 K and 0.001, respectively. The corresponding maximum RMSEs are about 1.1 K and 0.004. As the frequency increases, the RMSEs of the NN-FAPS increase slightly. One exception is at 23.8 GHz, probably because 23.8 GHz is located near the water vapor absorption line at 22.235 GHz.

To reduce the nonlinear error introduced by the linear approximation between radiance and BT, we introduced a Planck correction term for each channel, as shown in Fig. 3. It can be found that the linear approximation overestimates BT and the overestimation increases gradually with frequency (e.g., the overestimates can reach 0.8 K at 36.5 GHz), and the differences are almost negligible between different polarizations.

To better capture how NN-FAPSs predict the variations of the atmospheric parameters with respect to the input parameters, the SHAP values of NN-FAPS were calculated based on 5000 random training data and 1000 validation data. It can be found that the SHAP values of H polarization and $T^{\uparrow}$ are very close to the values of V polarization and $T^{\downarrow}$, respectively. Therefore, only the SHAP values of $T^{\downarrow}$ (Fig. 4) and Γ (Fig. 5) at V polarization are shown.

For the two frequencies where atmospheric effects are often assumed to be negligible, i.e., 6.925 and 7.3 GHz, it can be found that the contributions of NAST, TCWV, and TCIW to $T^{\downarrow}$ and Γ are almost negligible. However, as TCLW, TCRW, and TCSW increase, it is still necessary to consider the effects of clouds and precipitants on the BT at these two frequencies. Note that the color bars in Figs. 4 and 5 represent normalized feature values. Thus, the higher SHAP values of TCLW are caused by higher TCLW values than TCRW values, rather than TCLW having a higher contribution at the same value as TCRW.

For $T^{\downarrow}$, the SHAP values of all input parameters gradually increase with the increase of the corresponding input

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TABLE I

BIASES AND RMSES OF THE 36 NN-FAPSS ON TRAINING AND VALIDATION DATASETS CONSTRUCTED BASED ON RTTOV SIMULATION RESULTS. THE VALUES OUTSIDE/INSIDE THE BRACKETS REPRESENT THE PERFORMANCES ON THE TRAINING/VALIDATION DATASETS

Frequency (GHz)	Polarization	Downwelling BT (K)		Upwelling BT (K)		Transmissivity×100	
		Bias	RMSE	Bias	RMSE	Bias	RMSE
6.925	V	-0.03(-0.02)	0.37(0.41)	-0.02(-0.01)	0.36(0.41)	0.02(0.01)	0.17(0.19)
	H	-0.07(-0.05)	0.35(0.41)	-0.02(-0.01)	0.39(0.43)	-0.01(-0.01)	0.17(0.19)
7.3	V	0.03(0.04)	0.37(0.44)	-0.01(0.01)	0.37(0.43)	0.03(0.02)	0.18(0.22)
	H	0.08(0.10)	0.40(0.47)	0.07(0.08)	0.39(0.45)	0.04(0.03)	0.21(0.21)
10.65	V	0.17(0.18)	0.59(0.68)	-0.01(0.01)	0.62(0.68)	0.02(0.01)	0.28(0.29)
	H	-0.18(-0.16)	0.61(0.71)	-0.14(-0.12)	0.66(0.65)	-0.03(-0.04)	0.26(0.30)
18.7	V	-0.11(-0.09)	0.86(0.88)	0.04(0.05)	0.87(0.87)	-0.07(-0.08)	0.34(0.35)
	H	-0.04(-0.02)	0.89(0.92)	-0.01(0.03)	0.88(0.86)	-0.06(-0.07)	0.35(0.37)
23.8	V	0.06(0.06)	0.62(0.64)	0.09(0.08)	0.61(0.64)	-0.04(-0.04)	0.28(0.27)
	H	-0.09(-0.08)	0.66(0.64)	0.01(0.01)	0.63(0.61)	0.03(0.02)	0.27(0.28)
36.5	V	-0.03(-0.01)	1.05(1.07)	-0.04(-0.02)	1.07(1.08)	0.04(0.03)	0.42(0.42)
	H	0.24(0.25)	1.07(1.10)	-0.08(-0.07)	1.09(1.11)	-0.03(-0.05)	0.43(0.43)

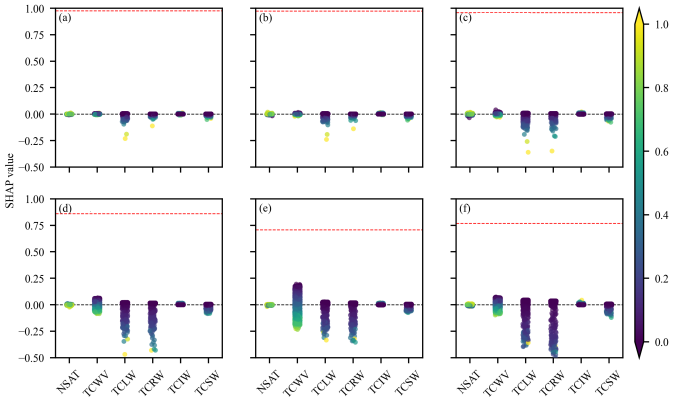

Fig. 5. SHAP values of transmissivity at V polarization for different frequencies. The color bar represents the normalized feature value. The red dashed line represents the baseline value of NN-FAPS (i.e., model outputs when all input parameters are 0). (a) 6.925, (b) 7.3, (c) 10.65, (d) 18.7, (e) 23.8, and (f) 36.5 GHz.

parameter values, but an opposite phenomenon occurs on the TCIW. The reason may be that suspended ice particles tend to be present in the colder upper atmosphere. Compared with 18.7 and 36.5, 23.8 GHz has lower TCLW and TCRW SHAPs but higher TCWV SHAPs. We would like to emphasize here that at 23.8 GHz, although the corresponding SHAP value is ~ -50 when the TCWV is close to 0, this does not mean that the water vapor contributes negatively to $T^{\downarrow}$. The reason is that there exists a baseline (i.e., model outputs when all input parameters are 0) when calculating the SHAP value. In other words, a negative SHAP value does not represent a negative contribution. However, when the SHAP value increases with the input parameter, the parameter has positive contributions to the model outputs. For liquid water condensate, the maximum values of TCLW and TCRW are 11.38 and 4.84 kg/m^2, respectively, in which case the SHAP value of TCRW is higher than that of TCLW at 18.7, 23.8, and 36.5 GHz.

For Γ, the baseline values decrease with frequency. Also, at 23.8 GHz, the baseline value of Γ is lower than 36.5 GHz due to higher water vapor absorption. Among the six input parameters, the contribution of NSAT is almost negligible; TCIW has a minor positive contribution; and TCWV, TCLW, TCRW, and TCSW have negative contributions, especially when the frequency is greater than 18.7 GHz. Overall, the contributions of TCLW and TCRW are more significant compared to TCWV and TCSW.

V. SSE UNDER RAINFALL CONDITION

After strict spatial–temporal matching and removal of land-containing pixels, there were 155 eligible AMSR2 pixels. The spatial distribution of spatial–temporal matching results between OceanRAIN and AMSR2 is shown in Fig. 6. The matched AMSR2 pixels are relatively evenly distributed overall. The numbers of pixels with rainfall percentages greater than 90% and convective rainfall percentages greater than 0 are 33 and 36, respectively. Note that convective rainfall in this study refers to the presence of convective rainfall events rather than purely convective rainfall within the pixel. Then, the SSE under rainfall conditions was calculated by the processes described in Step III. The rainfall-induced SSE change (Δemissivity) for V and H polarizations is shown in Figs. 7 and 8, respectively. In addition, considering that the SSE varies with WS and SST, distributions of Δemissivity with WS and SST are also shown.

Theoretically, it should be with the same WS and SST when analyzing the relationship between PERR and Δemissivity. However, it is very difficult to find enough samples with the same WS and SST (even in a small interval) due to the limitation of the number of samples. Therefore, only the qualitative relationships between Δemissivity and the three variables were analyzed based on the entire sample. It should be noted that PERR and WS/SST are little correlated, and together with the fact that WS/SST is relatively uniformly distributed (not shown here), the qualitative relationship between Δemissivity and PERR obtained on the entire sample is plausible. Overall, the impact of rainfall on SSE increases with frequency and PERR. At the same frequency, the SSE change of H polarization is greater than that of V polarization. When PERR is less than 3 mm/h, the maximum values of Δemissivity for the selected frequencies at V/H polarization are 0.06/0.11, 0.07/0.13, 0.16/0.27, 0.23/0.46, 0.21/0.41, and 0.27/0.68. Therefore, considering the change of SSE under rainfall conditions is crucial for all-sky applications of MW radiometers. A closer look reveals that the patterns of SSE

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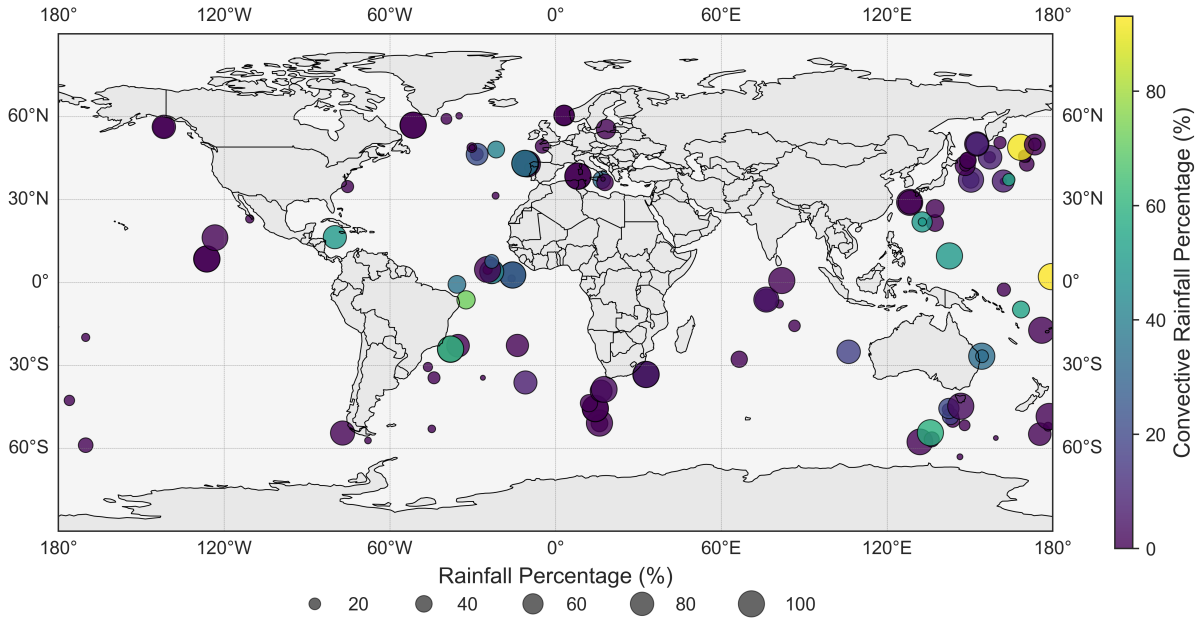

Fig. 6. Spatial distribution of spatial–temporal matching results between OceanRAIN samples and AMSR2 pixel. The sizes of the symbol represent the rainfall percentages of OceanRAIN samples within the AMSR2 pixel, and the colors represent the convective rainfall percentages.

change with PERR are quite different in the case of stratiform and convective rain. For stratiform rain (i.e., spatially homogeneous), the SSE increases rapidly with PERR. However, larger PERRs for convective rain are required to reach the stratiform-like SSE change values. The reason may be that the contribution of convective rainfall events to PERR is higher than that to SSE with the same percentage of convective rain.

The rainfall-induced SSE changes tend to be larger at lower WS, and this phenomenon is more pronounced at low-frequency and convective rainfall conditions. In TSM, WS affects the sea roughness spectrum, thereby affecting the scattering characteristics of MW. Generally, WS at different heights can be converted using friction velocity as a medium. However, the falling raindrops cause a downdraft, which is deflected horizontally at the sea surface and follows the momentum transfer equation. Note that the 10-m WS of OceanRAIN was converted from the higher altitudes (i.e., instruments installation heights) using a logarithmic wind profile [43]. Therefore, the 10-m WS of OceanRAIN may need to be revised to characterize the roughness spectrum caused by sea surface wind under rainfall conditions. Compared with stratiform rainfall, convective rainfall has larger raindrop sizes and terminal velocities, resulting in stronger local winds over the sea surface. In addition, raindrops hitting the sea surface induce ring waves, and SST affects the absorption and scattering of sea surface waves [44]. Therefore, the rainfall-induced SSE change is positively correlated with SST to a certain extent, which is consistent with the phenomenon in Figs. 7 and 8. Note that stratiform and convective rainfall rates are not uniformly distributed but have more low rainfall samples. The results of the analysis on the entire sample showed that Δemissivity was negatively correlated with WS and positively correlated with SST. Assuming that more high rainfall samples exist to make the PERR uniformly distributed, the negative correlation between Δemissivity and WS can be stronger. Similarly, the positive correlation between Δemissivity and SST can be stronger.

VI. SSE Error Analysis

As described in Section III-B, the three atmospheric parameters, Planck correction term, BT observed by the MW sensor, and surface temperature have essential impacts on the accuracy of SSE estimates. Therefore, it is necessary to quantify the error sources of the SSE under rainfall conditions. In this study, the SSE error analysis includes the SSE errors due to the fitting errors of NN-FAPSs and the Planck correction term based on (7) and the SSE errors in the real situation based on (9).

A. SSE Error Due to Fitting Errors

The SSE errors induced by the fitting errors of NN-FAPSs and Planck correction term are based on the results of NN-FAPS on the validation set and are shown in Fig. 9. Considering that extreme cases may lead to anomalies in the emissivity calculations, the extreme cases were excluded based on the three-standard deviations (3-σ) criterion. For example, when the true Γ is 0.02 and the NN-FAPS prediction is 0.01, the absolute difference is small, but this may lead to a significant overestimation of SSE. First, we calculated the means and STDs of the three atmospheric parameters on the validation set, and then, any sample whose absolute value of deviation from the mean was greater than 3 $\times$ STD that would not be used for counting SSE errors. It can be found from Fig. 9 that the SSE error ranges are below 0.001 $\pm$ 0.002 for 6.925, 7.3, and 10.6 GHz. For 18.7, 23.8, and 36.5 GHz, the corresponding ranges are below 0.01 $\pm$ 0.03. Overall, the SSE errors induced by the fitting errors of NN-FAPSs and the Planck correction term are acceptable, especially for the low frequency (i.e., $<$ 10.65 GHz). However, the maximum values

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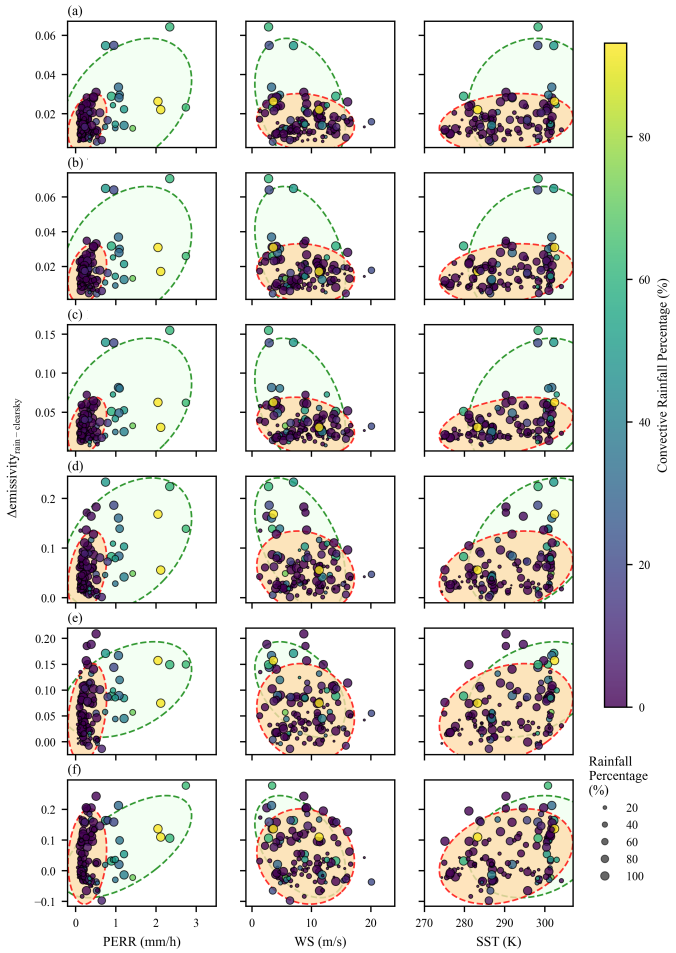

Fig. 7. Distribution of rainfall-induced SSE changes at V polarization with adjusted PERR, WS, and SST. The orange and green ellipses represent the confidence ellipses (2 × STD) of stratiform rainfall and convective rainfall, respectively. (a) 6.925, (b) 7.3, (c) 10.65, (d) 18.7, (e) 23.8, and (f) 36.5 GHz.

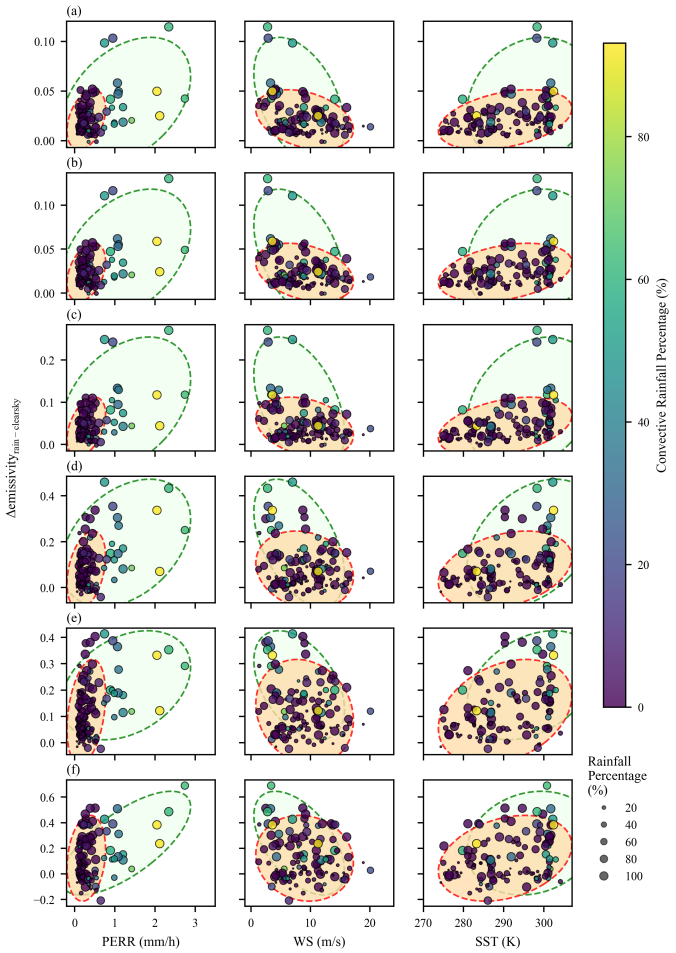

Fig. 8. Distribution of rainfall-induced SSE changes at H polarization with adjusted PERR, WS, and SST. The orange and green ellipses represent the confidence ellipses (2 × STD) of stratiform rainfall and convective rainfall, respectively. (a) 6.925, (b) 7.3, (c) 10.65, (d) 18.7, (e) 23.8, and (f) 36.5 GHz.

of SSE errors for the six frequencies are 0.02, 0.02, 0.04, 0.10, 0.45, and 0.55, indicating that despite the criterion filtering, the SSE errors are still completely unacceptable in specific conditions.

To better illustrate the effect of constructed NN-FAPS on the SSE error under different atmospheric conditions, the distributions of the six NN-FAPS inputs for different SSE error ranges were counted. Here, we only show the input distributions for 6.925 and 36.5 GHz (Fig. 10) since they are the least and most sensitive to extreme atmospheric conditions among the six frequencies selected. It can be found that the distribution of SSE error due to NN-FAPS is almost independent of the distribution of NSAT. For the other five inputs, the increasing input values lead to an increase in SSE error. At the same time, the threshold values of the input corresponding to different SSE error ranges gradually decrease with increasing frequency. For example, the TCRW of the samples with $\left|\Delta\varepsilon - \overline{\Delta\varepsilon}\right| <$ STD lies mainly below ~1.3 kg/m^2 at 6.925 GHz, while the corresponding values are ~0.6 kg/m^2 at 36.5 GHz. In addition, the most pronounced threshold decreases occur in TCLW and TCRW, which corresponds to the higher SHAP values of the two inputs in Figs. 4 and 5.

B. SSE Error for Matched Samples

According to (9), in addition to the fitting errors of the NN-FAPSs, it is necessary to consider the SSE error caused by the NN-FAPS input parameters and the measurement errors of AMSR2 BT and SST for the matched samples. The errors given for different input parameters are shown in Table II. For NAST and SST, the errors were the STDs of the corresponding parameters for all OceanRAIN samples within the AMSR2 pixel. The STD ranges for NAST and SST were 0.02–1.99 and 0.01–0.32 K, respectively. For the input parameters from ERA5, relative errors were used considering that it is unreasonable to give fixed errors for different atmospheric conditions. The relative errors for TCWV and TCLW were derived from previous studies [45]. In addition, the relative errors for TCIW and TCSW were set to 50% due to their low contribution and low content on the matched samples (AMSR2 pixels with snowfall were removed during the sample matching).

The SSE error caused by the fitting errors of NN-FAPS and Planck correction term and the input parameter errors are given in Table III. For 6.925, 7.3, and 10.65 GHz, the SSE errors (both $\Delta\varepsilon$ and error from different sources) are below 0.01 ± 0.01, which indicates that the simulated SSEs at these

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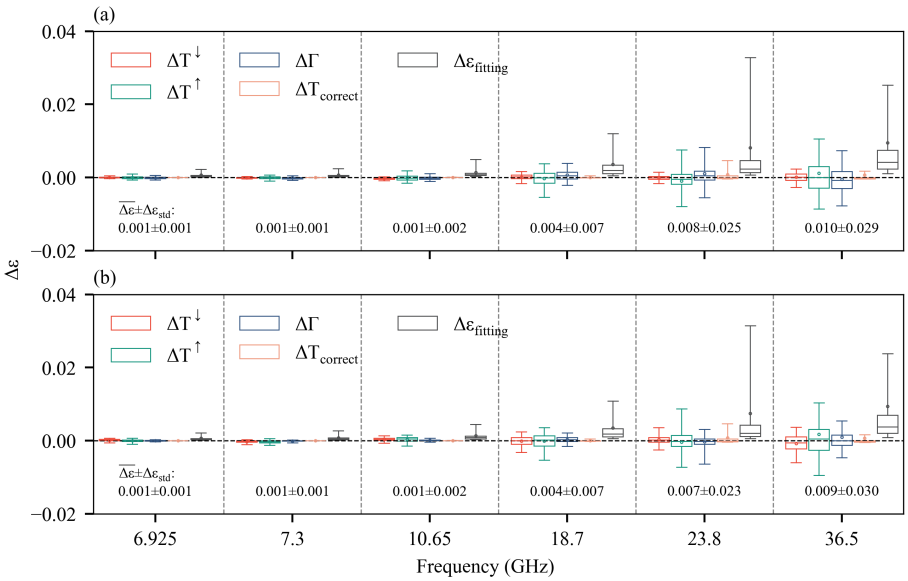

Fig. 9. Emissivity errors induced by the fitting errors of NN-FAPSs and Planck correction term at (a) V polarization and (b) H polarization. Different colored boxes represent different atmospheric parameters (i.e., $T^{\uparrow}$, $T^{\downarrow}$, Γ, and T_{correct}).

TABLE II

INPUT PARAMETER ERRORS OF USED AMSR2 BT, OCEANRAIN, AND ERA5. THE STD REPRESENTS THAT ERRORS WERE THE STDS OF THE CORRESPONDING PARAMETERS FOR ALL OCEANRAIN SAMPLES WITHIN THE AMSR2 PIXEL

Dataset	Parameter	Error	Reference
AMSR2 BT	6.925 GHz	0.34 K	[46]
	7.3 GHz	0.43 K	[46]
	10.65 GHz	0.7 K	[46]
	18.7 GHz	0.7 K	[46]
	23.8 GHz	0.6 K	[46]
	36.5 GHz	0.7 K	[46]
OceanRAIN	NSAT	*std* K	-
	SST	*std* K	-
	PERR	0.19 mm/h	[39]
ERA5	TCWV	25%	[45]
	TCLW	55%	[45]
	TCIW	50%	-
	TCSW	50%	-

three frequencies are little affected by the atmosphere. For 18.7, 23.8, and 36.5 GHz, the simulated SSEs are strongly influenced by the accuracy of atmospheric parameters (e.g., the SSE error of 23.8 GHz at H polarization can reach –0.2). Among the six sources contributing to SSE errors, the measurement errors of BT and SST have small contributions to the SSE errors. $\Delta T^{\uparrow}$, $\Delta T^{\downarrow}$, and $\Delta\Gamma$ due to input parameter errors have more significant effects on the SSE than the fitting error and the measurement errors of BT and SST. As for the fitting error, it is negligible when the frequency is less than 10.65 GHz. However, when the frequency is greater than 18.7 GHz, the fitting error shows a nonnegligible increase with frequency, but the magnitude is still smaller than the SSE error due to $\Delta T^{\uparrow}$, $\Delta T^{\downarrow}$, and $\Delta\Gamma$. Note that the trends of $T^{\uparrow}/T^{\downarrow}$ and Γ are opposite (i.e., higher $T^{\uparrow}/T^{\downarrow}$ corresponds to lower Γ) under different atmospheric conditions, which leads to opposite contributions to SSE. Therefore, biases of SSE from different sources can offset each other to a certain extent. This is also the reason why the bias of $\Delta\varepsilon$ is smaller than that of SSE caused by $\Delta T^{\uparrow}$ and $\Delta\Gamma$. In summary, although there are errors in the simulated SSEs, the errors are significantly smaller than the calculated SSE changes caused by rainfall, which indicates that rainfall does have a nonnegligible effect on the SSE.

Additionally, an exception appears in Figs. 7 and 8 that the rainfall-induced SSE changes appear negative on some samples at 18.7, 23.8, and 36.5 GHz (i.e., the rainfall SSE is lower than the clear-sky SSE). Based on the results of the SSE error analysis, the main reason for this phenomenon is that ERA5 may overestimate TCLW under rainfall conditions, but this overestimation may not be systematic.

VII. FITTING MODEL FOR THE RELATIONSHIP BETWEEN RAINFALL AND CLEAR-SKY EMISSIVITY

Although the calculated SSE of the matched samples was affected by the accuracies of the four ERA5 parameters (i.e., TCWV, TCLW, TCIW, and TCSW), the results of the error analysis based on the matched samples showed that $\Delta\varepsilon$ at 6.925, 7.3, and 10.65 GHz was below 0.01 ± 0.01. Therefore, we built polynomial regression models for these three frequencies to quickly calculate the SSE under rainfall conditions with

$$\begin{aligned}\varepsilon_{\text{rain,pred}} = {} & a_0 + a_1\varepsilon_{\text{clear-sky}} + a_2\text{RR} + a_3\text{SST} + a_4\text{WS} \\ & + a_5\varepsilon^2_{\text{clear-sky}} + a_6(\varepsilon_{\text{clear-sky}} \cdot \text{RR}) \\ & + a_7(\varepsilon_{\text{clear-sky}} \cdot \text{SST}) + a_8(\varepsilon_{\text{clear-sky}} \cdot \text{WS}) \\ & + a_9(\text{RR}^2) + a_{10}(\text{RR} \cdot \text{SST}) + a_{11}(\text{RR} \cdot \text{WS}) \\ & + a_{12}(\text{SST}^2) + a_{13}(\text{SST} \cdot \text{WS}) \\ & + a_{14}(\text{WS}^2) \end{aligned} \tag{11}$$

where $\varepsilon_{\text{rain,pred}}$ is SSE under rainfall conditions predicted by the polynomial regression models, $\varepsilon_{\text{clear-sky}}$ is the SSE calculated by the SURFEM-Ocean, RR is the rain rate in

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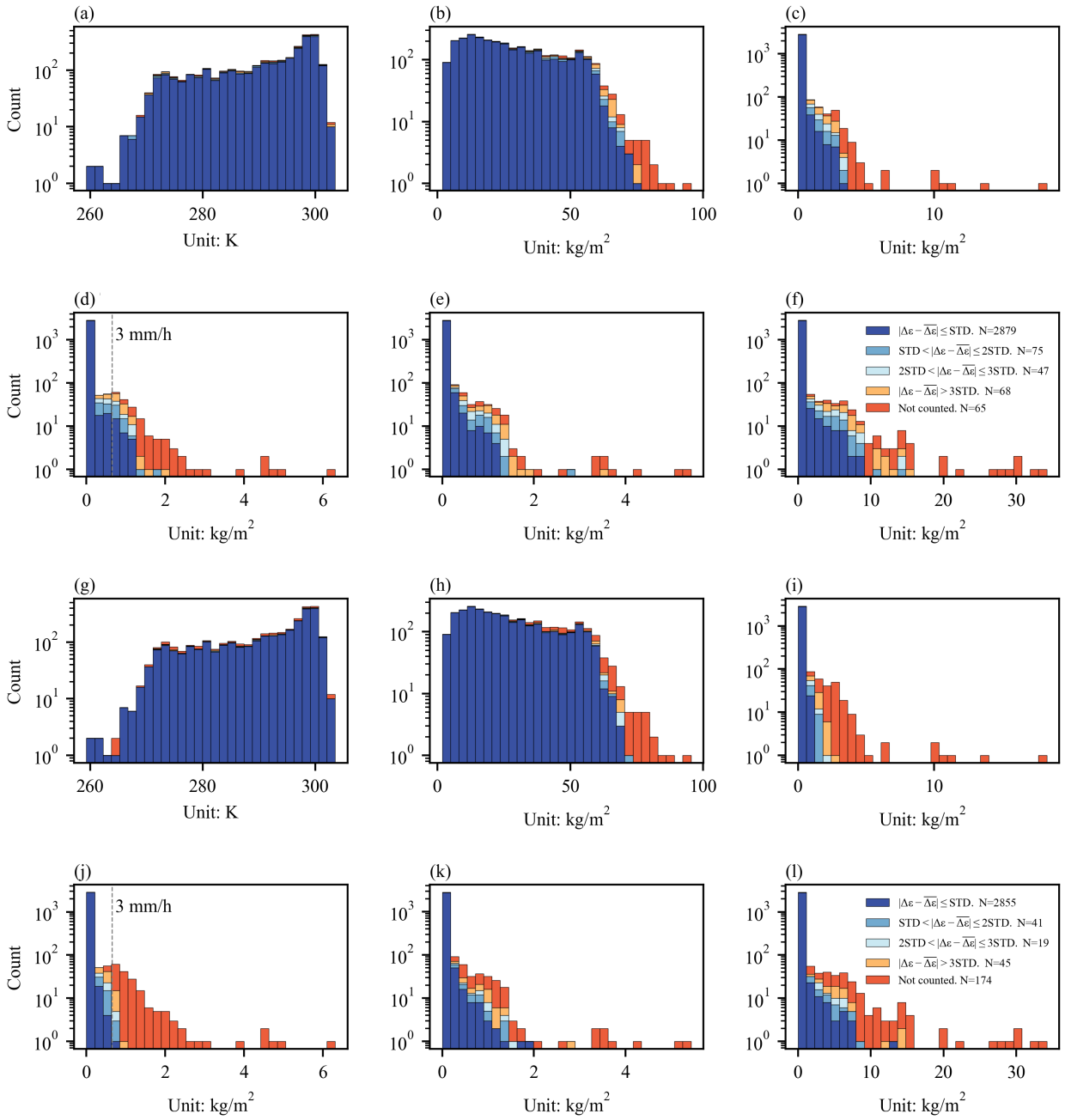

Fig. 10. Distributions of (a) and (g) NSAT, (b) and (h) TCWV, (c) and (i) TCLW, (d) and (j) TCRW, (e) and (k) TCIW, and (f) and (l) TCSW for different SSE error ranges at the V polarization of (a)–(f) 6.925 and (g)–(l) 36.5 GHz. The gray vertical dashed line in (d) and (j) represents the TCRW corresponding to the pixel RR of 3 mm/h.

mm/h, SST is the sea surface temperature in K, WS is the wind speed at 10 m in m/s, and a_i ($0 \leq i \leq 14$) is the coefficient. The scatter plots of the calculated and predicted SSEs under rainfall conditions are shown in Fig. 11. The coefficients and fitting accuracy are shown in the Appendix and Table IV, respectively. At the same frequency and polarization, the bias of convective rainfall is significantly smaller than that of stratiform rainfall, but the opposite occurs for RMSE. Also, the fitting accuracies (both bias and RMSE) at H polarization are lower than those at V polarization. For frequencies below 10 GHz (i.e., 6.925 and 7.3 GHz), the RMSEs are below 0.01 and 0.02 at the V and H polarizations, respectively. As for 10.65 GHz, the corresponding RMSEs are below 0.25 and 0.4. It is believed that the constructed polynomial model, if used as the first guess of SSE in the assimilation model, can improve the accuracy of existing cloud and rainfall products.

VIII. Discussion

A. Influences Caused by Neglecting the Rainfall-Induced SSE Changes

Accurate MW SSE is a key parameter for various all-sky applications based on MW imagers. The results in Section V confirm that rainfall leads to higher SSE and this phenomenon becomes more obvious with increasing frequency. It is known

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TABLE III
SSE ERRORS CAUSED BY THE FITTING ERRORS AND THE INPUT PARAMETER ERRORS

Frequency	$\Delta\varepsilon_{\text{fitting}}$	$\Delta\varepsilon_{\Delta T\uparrow(\Delta\mathbf{X})}$	$\Delta\varepsilon_{\Delta T\downarrow(\Delta\mathbf{X})}$	$\Delta\varepsilon_{\Delta\Gamma(\Delta\mathbf{X})}$	$\Delta\varepsilon_{\Delta BT}$	$\Delta\varepsilon_{\Delta Ts}$	$\Delta\varepsilon$
6.925V	0.001±0.001	-0.003±0.001	-0.001±0.001	0.002±0.001	0.001±0.001	-0.001±0.001	-0.001±0.002
6.925H	0.001±0.001	-0.003±0.001	-0.002±0.001	0.001±0.001	0.001±0.001	-0.001±0.001	-0.002±0.002
7.3V	0.001±0.001	-0.003±0.002	-0.001±0.001	0.001±0.002	0.002±0.001	-0.001±0.001	-0.002±0.003
7.3H	0.001±0.001	-0.004±0.002	-0.003±0.001	0.001±0.001	0.002±0.001	-0.001±0.001	-0.003±0.003
10.65V	0.001±0.002	-0.009±0.004	-0.003±0.001	0.006±0.003	0.003±0.001	-0.002±0.001	-0.004±0.005
10.65H	0.001±0.002	-0.008±0.003	-0.005±0.002	0.003±0.001	0.003±0.001	-0.001±0.001	-0.008±0.005
18.7V	0.004±0.007	-0.049±0.018	-0.014±0.005	0.036±0.015	0.003±0.001	-0.002±0.001	-0.022±0.025
18.7H	0.004±0.007	-0.049±0.018	-0.026±0.008	0.024±0.013	0.003±0.001	-0.001±0.001	-0.045±0.025
23.8V	0.008±0.025	-0.166±0.091	-0.033±0.013	0.137±0.081	0.005±0.002	-0.002±0.001	-0.051±0.125
23.8H	0.007±0.023	-0.165±0.091	-0.059±0.023	0.109±0.072	0.005±0.002	-0.001±0.001	-0.105±0.120
36.5V	0.010±0.029	-0.134±0.058	-0.029±0.011	0.116±0.058	0.005±0.002	-0.002±0.001	-0.035±0.087
36.5H	0.009±0.030	-0.136±0.061	-0.050±0.020	0.091±0.057	0.005±0.002	-0.002±0.001	-0.082±0.091

TABLE IV
ACCURACY METRICS OF THE POLYNOMIAL REGRESSION MODELS

	6.925 GHz				7.3 GHz				10.65 GHz			
	V		H		V		H		V		H	
	stratiform	convective	stratiform	convective	stratiform	convective	stratiform	convective	stratiform	convective	stratiform	convective
bias	**0.001**	**≪0.001**	**0.003**	**≪0.001**	**0.001**	**≪0.001**	**0.003**	**≪0.001**	**0.002**	**≪0.001**	**0.005**	**≪0.001**
RMSE	**0.006**	**0.008**	**0.011**	**0.015**	**0.006**	**0.009**	**0.011**	**0.018**	**0.012**	**0.021**	**0.023**	**0.040**

that rainfall and cloud liquid water heat the MW radiation emitted by the ocean (due to the low SSE). Ignoring the SSE change caused by rainfall (i.e., underestimating the SSE) will lead to an underestimation of the MW radiation emitted by the ocean, and more heating is needed to ensure that the radiation transferred to the top of atmosphere (TOA) is consistent with the observation, which may lead to an overestimation of the rainfall content and cloud liquid water content. In addition, considering the effects of local wind caused by rainfall and SST on the dielectric constant, overestimation of rainfall is more likely to occur under conditions of low WS and high SST.

In addition, it should be noted that this overestimation not only appears in satellite products but also affects data assimilation and reanalysis. For example, the all-sky radiances of the TRMM Microwave Imager (TMI) and some channels of the Special Sensor Microwave Imager/Sounder have been assimilated by the Integrated Forecasting System Cycle 41r2 to produce the ERA5 dataset, and the forward radiative transfer model used is RTTOV-11, which calculates the MW SSE based on FASTEM. Therefore, ERA5 also has the possibility of overestimating rainfall and cloud liquid water content. Nogueira [47] compared ERA5 and Global Precipitation Climatology Project (GPCP) rainfall over the past 40 years and found that ERA5 tended to overestimate rainfall over the ocean and this phenomenon was more significant over most tropical oceans (1 mm/day), which is consistent with our inference that overestimation tends to occur at high SST. This suggests that ignoring rainfall-induced SSE changes may be one of the reasons for the overestimation of rainfall in ERA5. Therefore, taking rainfall-induced SSE changes into account is critical in improving the accuracy of satellite-based MW radiometer products as well as assimilation and reanalysis products.

B. Future Outlook

In this study, the SSEs under rainfall conditions were calculated based on the RTE with the satellite passive MW observations, in situ ship measurements, and ERA5. However, this method has some drawbacks: 1) its accuracy is limited by the accuracy of the input parameters, especially TCLW and TCRW; 2) the SSE calculated by this method is difficult to extend to other observation geometries and other nonobservation frequencies; and 3) for rough sea surfaces, this method cannot correct the sea surface reflectivity (SSR) by anisotropic

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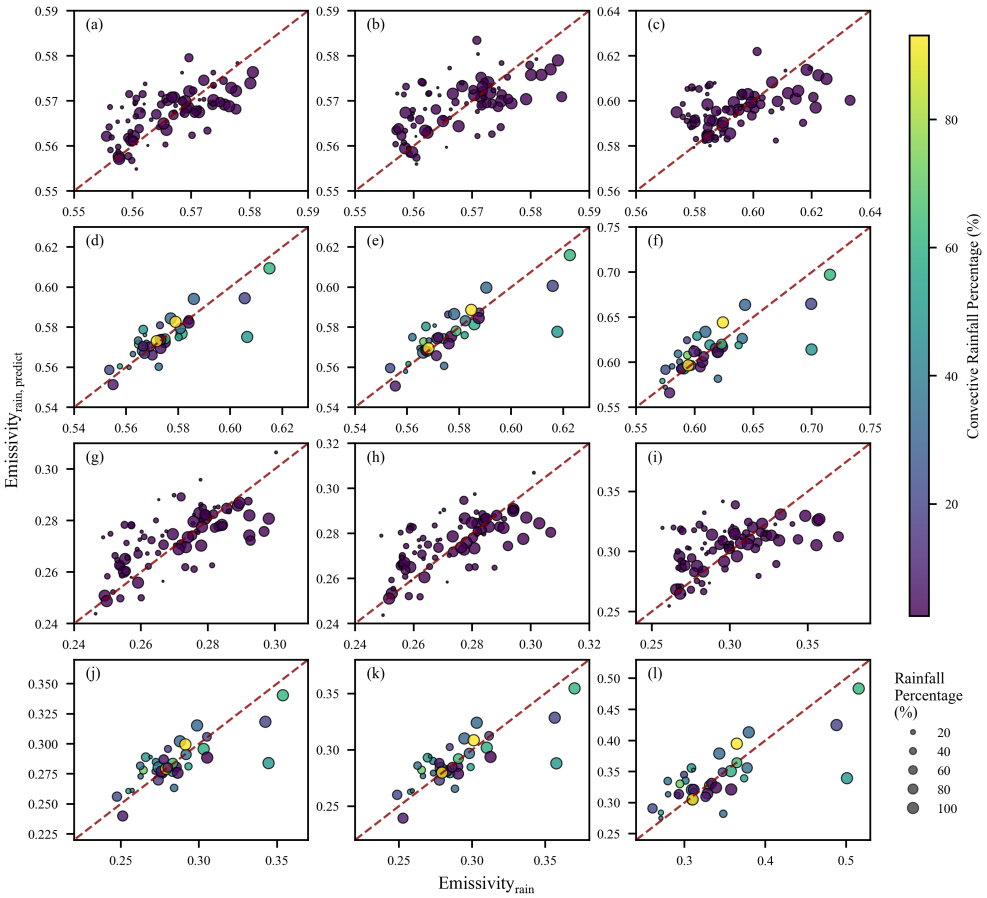

Fig. 11. Scatter plots of the calculated and predicted SSEs under rainfall conditions for (a), (d), (g), and (j) 6.925; (b), (e), (h), and (k) 7.3; and (c), (f), (i), and (l) 10.65 GHz.

downwelling radiation, i.e., the SSR must always be assumed to be 1 minus SSE. Beyond the accuracies of the input parameters, some key assumptions in radiative transfer modeling for rainfall scenarios may introduce systematic uncertainties. The spherical raindrop shape assumption can lead to errors in polarized scattering properties. The commonly used raindrop size distribution parameterization (i.e., Marshall–Palmer model [48]) puts high emphasis on smaller drops, resulting in stronger backscattering compared with other raindrop size distribution models (e.g., Wang2016 model [49]). Additionally, oversimplified treatments of multiple scattering processes may underestimate radiation path enhancement under high rainfall rates. Therefore, MW all-sky SSE models based on physical models should be developed in the future, such as coupling gravity waves and capillary waves caused by sea surface wind and ring waves and local sea surface wind caused by rainfall. Once the physics-based all-sky SSE model is fully established, the current overestimation in existing rainfall products and reanalysis may be significantly alleviated. Subsequently, this can improve numerical weather prediction and climate models, thereby advancing the scientific community's understanding and predictive capabilities concerning climate change trends and extreme weather events.

IX. Conclusion

SSE is a key input parameter for satellite-based MW radiometer products as well as assimilation and reanalysis products, and clear-sky SSE is often applied directly to rainfall scenarios currently. In this study, SSEs under rainfall conditions were inferred from satellite passive MW observations and in situ ship measurements (OceanRAIN) based on RTE.

First, the NN-FAPSs for the AMSR2 frequencies (except 89.0 GHz) were developed to quickly calculate the atmospheric parameters, i.e., atmospheric upwelling BT ($T^{\uparrow}$), atmospheric downwelling BT ($T^{\downarrow}$), and transmissivity (Γ), in the RTE based on the simulation results of RTTOV-SCATT. The input parameters of NN-FAPS include NSAT, TCWV, TCLW, TCRW, TCIW, and TCSW. In addition, a Planck correction term was introduced to reduce the nonlinear error caused by the Rayleigh–Jeans approximation. For $T^{\uparrow}/T^{\downarrow}$ and Γ, the maximum RMSEs of NN-FAPSs occur at 36.5 GHz and are $\sim$1.1 K and 0.004, respectively. SHAP values of NN-FAPSs showed that TCLW and TCRW have more important effects on the three atmospheric parameters. Also, TCLW and TCRW contribute positively to $T^{\uparrow}$ and $T^{\downarrow}$, but negatively to Γ. For NSAT, TCWV, and TCSW, the overall SHAPs were small but still increased with frequency (at 23.8 GHz, the SHAP increases rapidly with TCWV). As for TCIW, its SHAP value is also small but contributes negatively to $T^{\uparrow}$ and $T^{\downarrow}$ and positively to Γ.

With the NN-FAPSs and the Planck correction term, the SSE under rainfall conditions was calculated from the spatial–temporal matching results between OceanRAIN samples and AMSR2 pixel. During the spatiotemporal

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TABLE V
FITTING COEFFICIENTS (a_i) OF THE POLYNOMIAL REGRESSION MODELS

	6.925 GHz				7.3 GHz				10.65 GHz			
	V		H		V		H		V		H	
	stratiform	convective	stratiform	convective	stratiform	convective	stratiform	convective	stratiform	convective	stratiform	convective
a_0	24.61045	73.51767	16.96759	-9.53967	48.41414	88.3964	16.88286	-16.8501	57.32246	297.1978	16.4846	-45.2249
a_1	-84.1426	-497.475	-187.019	-109.744	-178.28	-533.977	-178.992	-79.2785	-150.443	-1305.46	-113.096	-123.188
a_2	4.108845	1.822322	3.297279	1.711465	4.100141	0.960895	3.284673	1.606338	9.145653	2.438259	10.194	1.848786
a_3	-0.01403	0.404111	0.023602	0.136128	-0.00131	0.378405	0.018439	0.161058	-0.11362	0.412232	-0.03829	0.372298
a_4	0.14194	1.004611	0.592685	0.656761	0.233208	1.026714	0.579894	0.629787	0.212146	2.4892	0.529464	1.318247
a_5	70.55406	716.5116	423.6253	800.9248	160.8329	725.7011	382.2378	794.3952	92.38538	1383.199	117.4621	1157.925
a_6	-7.82331	-1.62335	-13.3343	-0.02502	-7.46734	0.23586	-12.6919	1.228166	-14.2878	-0.80364	-31.7046	6.099742
a_7	0.036364	-0.92907	0.014115	-0.83059	0.021876	-0.85736	0.04803	-0.92494	0.181029	-0.73653	0.256334	-1.34574
a_8	-0.25091	-2.30726	-2.71381	-5.5876	-0.43765	-2.29283	-2.51287	-5.77708	-0.28259	-4.695	-1.34613	-9.99806
a_9	-0.0114	-0.01511	-0.01709	-0.02574	-0.0088	-0.01677	-0.06499	-0.02892	-0.11536	-0.03594	-0.16087	-0.05947
a_{10}	0.000775	-0.00285	-0.00035	-0.00528	0.000208	-0.00337	-0.00065	-0.00585	-0.00344	-0.00606	-0.0081	-0.01006
a_{11}	-0.00031	-0.00619	0.018268	-0.01138	-0.00192	-0.00754	0.016643	-0.01525	3.75E-05	-0.01434	0.06029	-0.03875
a_{12}	-9.98E-06	0.000178	-4.67E-05	0.000101	-1.85E-05	0.000159	-5.08E-05	9.78E-05	2.47E-05	8.86E-06	-3.07E-05	-7.03E-05
a_{13}	-2.07E-05	0.000841	3.49E-05	0.002014	1.79E-05	0.000742	-7.09E-05	0.002254	-0.00019	0.000346	-0.00084	0.003243
a_{14}	0.000169	0.001568	0.003936	0.009392	0.000251	0.001801	0.003731	0.010074	0.000176	0.003919	0.002557	0.020363

matching process, along-track averaging and two track-to-area adjustments were used to resolve the inconsistency between the spatial scales of OceanRAIN point measurements and AMSR2 pixels. The results showed that rainfall causes an increase in SSE (e.g., can up to 0.2 at 10.65 GHz under low WS and high SST), and this phenomenon is more pronounced at low WSs and high SST, corresponding to the effects of rainfall-induced local wind and SST on the dielectric constant. In addition, with the same PERR, stratiform rainfall induces a larger SSE change compared to convective rainfall. The results of the error analysis showed that at 6.925, 7.3, and 10.65 GHz, the accuracy of the SSE is less affected by the accuracy of the input parameters as well as the fitting errors of the NN-FAPSs and the Planck correction term ($\Delta\varepsilon$ was below 0.01 ± 0.01). However, the SSE error cannot be ignored at 18.7, 23.8, and 36.5 GHz, and the maximum SSE error can reach -0.11 ± 0.12. However, it is important to note that despite the very large SSE error, the magnitude of the SSE error is still smaller than the calculated SSE change due to rainfall. Therefore, the SSE errors due to input parameter errors and fitting errors cannot adequately explain the SSE changes under rainfall conditions. In summary, rainfall causes a nonnegligible increase in SSE, which becomes more pronounced with increasing frequency. Ignoring the SSE change caused by rainfall (i.e., underestimating the SSE) may lead to an overestimation of satellite-based and reanalysis rainfall products, which has been confirmed by Nogueira [47]. Finally, polynomial models for 6.925, 7.3, and 10.65 GHz were constructed that can provide the first guess of SSE under rainfall conditions for the relevant satellite products and assimilation models, thereby improving the accuracy of cloud and rainfall products.

APPENDIX

The fitting coefficients of the polynomial regression models are shown in Table V.

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